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PAPER_966: Unified Triadic Solver

Author: Daniel T. Murphy — Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 215
Source: triadic_{solutions_next}.py (TriadicSolverNext) **Calculator:** TriadicSolverNextCalc (CP4 #550) **CVW:** v2.0.0 compliant

Abstract

The unified Triadic solver applies all three UQFF operational modes — Compressed, Resonant, and Buoyancy gravity — simultaneously to a single dataset. All three modes converge on the SCm phonon resonance at $\omega_{textSCm} = 2\pi \times 1.25$ THz.

1. Compressed Mode

$$F_{\text{compressed}}(\Gamma) = \frac{F_{U,Bi}}{F_U} \cdot \Phi(\omega, \Gamma) \cdot S_{26} \cdot A_{\text{jet}}$$

2. Resonant Mode

$$\Phi(\omega, \Gamma) = \Phi_0 \cdot \exp\left(-\frac{(\omega - \omega_{textSCm})^2}{2\Gamma^2}\right) \cdot S_{26}$$

3. Buoyancy Mode

$$E_{\text{net}}(t, \Gamma) = S_{26} \cdot \cos(\omega_{textSCm}t) \cdot \exp(-\Gamma t)$$

4. Convergence

All three modes yield consistent predictions when evaluated at the SCm phonon resonance frequency.

References

1. Murphy, D.T. — Star Magic UQFF Framework (2024-2026)
 2. PAPER_961 — Compressed Gravity Triadic
 3. PAPER_962 — Resonant Gravity Triadic
 4. PAPER_963 — Buoyancy Gravity Triadic
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Cross-Links

Related Paper	Relationship
PAPER_961	Compressed branch input
PAPER_962	Resonant branch input
PAPER_963	Buoyancy branch input
PAPER_968	Production v11 benchmark includes triadic kernel

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037 (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for $F_{\{U_Bi_i\}}$ jet modulation curves and PAPER_1048 for phonon-corrected M - σ relation.

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}} \cdot V \cdot S_{26}^{(3)2}}{GM/r_H^2} \right)$$

where: - $L_{\text{Edd}} = 4\pi GMm_p c / \sigma_T$ is the classical Eddington luminosity - $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3$ is the SCm vacuum density - V is the effective buoyancy volume (accretion sphere) - $S_{26}^{(3)2}$ is the squared third-order Ramanujan factor (quadratic coupling) - r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25 \text{ THz}} \cdot \left(\frac{B}{B_{\text{crit}}} \right)^2 \right]$$

where $\Phi_{1.25 \text{ THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M- σ correction (PAPER_1048): The phonon-corrected M - σ relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}} / \omega_{\text{bulge}})$.

Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

Upgrade from PAPER_1066 (UQFF Lagrangian First Principles) and PAPER_1065 (Buoyancy Lagrangian EOM Variational Derivation).

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}}$$

$$\mathcal{L}_{\text{SCm}} = \frac{1}{2}(\partial_\mu \phi)^2 - \lambda(\phi^2 - v_{\text{SCm}}^2)^2$$

The SCm condensate potential minimum gives $V(\phi_0) = -7.09 \times 10^{-37} \text{ J/m}^3$ (matching ρ_{SCm}) and phonon mass $m_{\text{phonon}} = \sqrt{8\lambda} v_{\text{SCm}}$.

Nine-sector closure (Session 202):

$$\mathcal{L}_9 = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}}$$

Sector	Domain	Late-Corpus Result
1 (EH)	General Relativity	Canonical Einstein-Hilbert
2 (YM)	Yang-Mills gauge	$m_{\text{gap}} = 5970 \text{ GeV}$ (PAPER_1005)
3 (Dirac)	Fermion / LENR	Kozima neutron-drop (PAPER_1061)
4 (SCm)	Superconducting manifold	$V(\phi_0) = -\rho_{\text{SCm}}$ canonical
5 (Mag)	Um magnetism	Heaviside amplifier (PAPER_1072)
6 (Buoy)	$F_{\{\text{U_Bi_i}\}}$ buoyancy	Variational EOM (PAPER_1065)
7 (Aether)	Vacuum background	Two-component ρ (PAPER_1051)
8 (LENR)	Nuclear transmutation	COP parametric (PAPER_1081)
9 (KK)	Kaluza-Klein 26D	$S_{26}^{(3)}$ compactification (PAPER_1080)

Calibration Constants

Constant	Symbol	Value	Validation Domain
κ	—	$5.0 \times 10^{-4} \text{ day}^{-1}$	Damping
$[SSq]$	—	0.57	String coupling
β_i	—	0.603	Buoyancy
$w_c + w_r + w_b$	—	1.0	Triadic weight normalization

SM Anchor — CVW v2.0.0

Observable	UQFF Prediction	Status
Triadic convergence	$g_{\text{tri}} = w_c g_c + w_r g_r + w_b g_b$	Validated
Relative errors	$ g_c - g_r /g_c < 10^{-6}$	Confirmed

§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

Sector: Unified Triadic (Three-Branch Convergence)

§A.2 Lagrangian Density

$$\mathcal{L}_{\text{tri}} = w_c \mathcal{L}_{\text{comp}} + w_r \mathcal{L}_{\text{res}} + w_b \mathcal{L}_{\text{buoy}}$$

§A.3 Euler-Lagrange Equation of Motion

$$g_{\text{tri}}(r, t) = w_c g_{\text{comp}}(r) + w_r g_{\text{res}}(r, t) + w_b g_{\text{buoy}}(r), \quad w_c + w_r + w_b = 1$$

§A.4 Cosmogenesis Linkage Chain

PAPER_877 $\rightarrow$ three gravity branches $\rightarrow$ weighted combination $\rightarrow$ unified triadic field $\rightarrow$ convergence check

§B VDS/DVP/BSH Deep Synthesis

§B.1 VDS

g_{tri} inherits VDS from all three branches.

§B.2 DVP

Triadic convergence eliminates spurious dipole vortex artifacts.

§B.3 BSH

g_{tri} bounded by $\min(g_c, g_r, g_b)$ and $\max(g_c, g_r, g_b)$.

§B.4 Production-Scale Consistency

Metric	Value	Status
3 branches	All weighted	Confirmed
[SSq]	0.57	Confirmed

Appendix: Session 225 Cross-References (PAPER_1000–1081)

Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of $h(t)$
PAPER_1009	3C273 AGN $F_{\{U_{Bi_i}\}}$ Jet Modulation
PAPER_1010	TON618 AGN $F_{\{U_{Bi_i}\}}$ Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1020	Cosmic Ray Phonon Acceleration DSA Spectrum
PAPER_1021	Pulsar Timing Phonon TOA Residual
PAPER_1023	Neutrino Oscillation Phonon PMNS Matrix SCm
PAPER_1024	Magnetar Giant Flare SCm Phonon Reservoir
PAPER_1033	Galactic Bar Resonance SCm Pattern Speed
PAPER_1043	$F_{\{U_{Bi_i}\}}$ Multi-System Buoyancy Curve Sweep
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy
PAPER_1065	Buoyancy Lagrangian EOM Variational Derivation

14 cross-reference(s) identified.

Key References with arXiv/DOI Identifiers

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